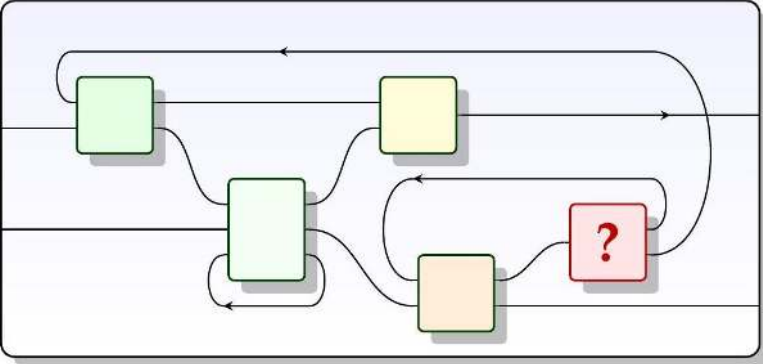
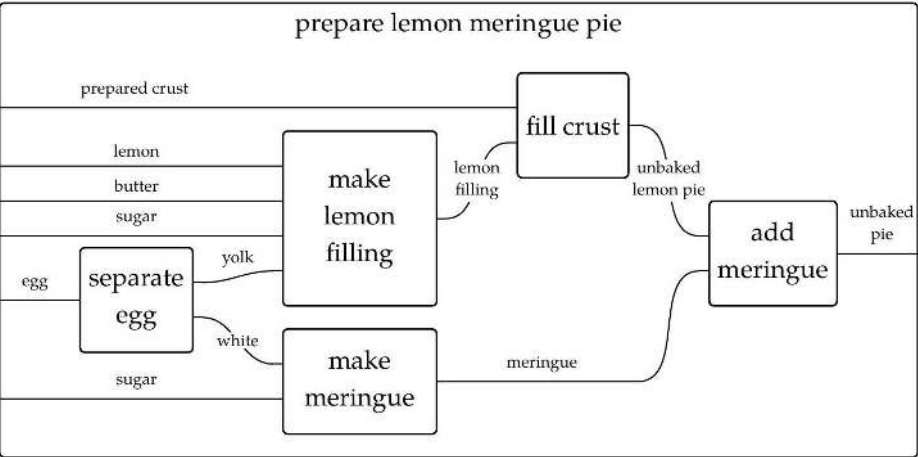
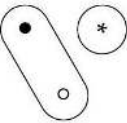
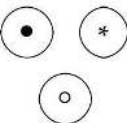
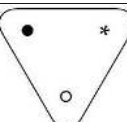
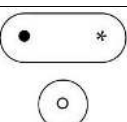
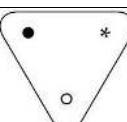


fong-spivak-seven-sketches: image evidence

Image regions

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fong-spivak-seven-sketches_4 DIA_0002	4	—	—	—	
fong-spivak-seven-sketches_15 DIA_0003	15	—	—	—	
fong-spivak-seven-sketches_15 DIA_0004	15	—	—	—	
fong-spivak-seven-sketches_15 DIA_0005	15	—	—	—	
fong-spivak-seven-sketches_15 DIA_0006	15	—	—	—	
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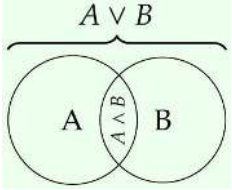
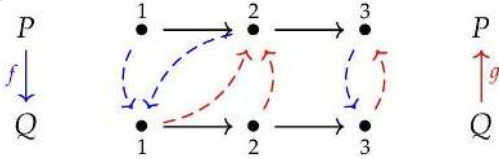
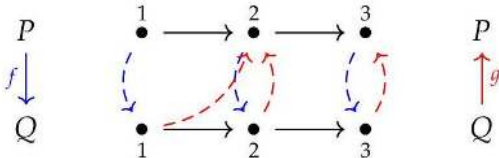
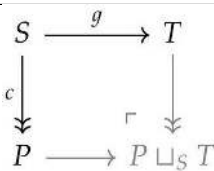
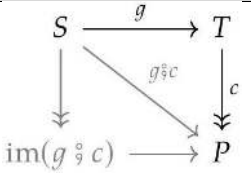
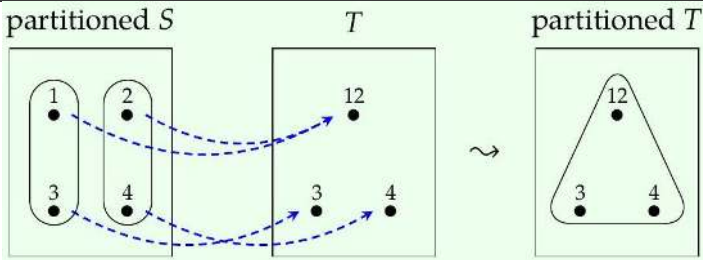
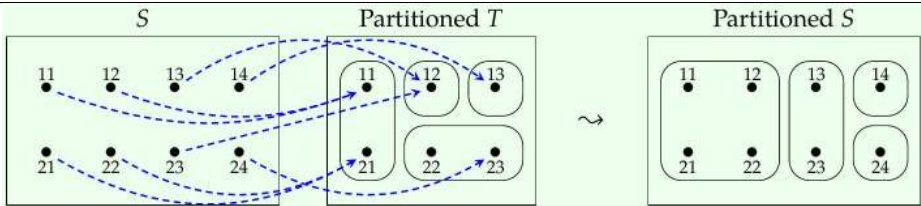
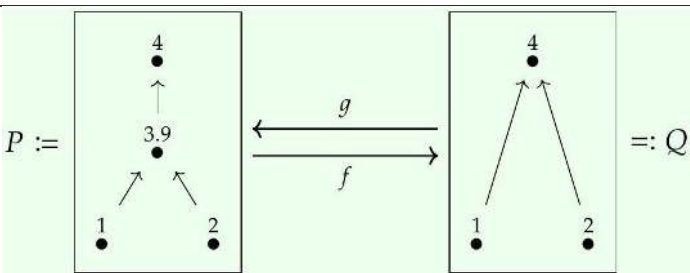
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fong-spivak-seven-sketches_16 DIA_0011	16	—	—	—	
fong-spivak-seven-sketches_16 DIA_0012	16	—	—	—	
fong-spivak-seven-sketches_16 DIA_0013	16	—	—	—	
fong-spivak-seven-sketches_17 DIA_0014	17	—	—	—	
fong-spivak-seven-sketches_17 DIA_0015	17	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_0016 DIA_0016	18	—	—	—	$\begin{array}{c} \text{true} \\ \uparrow \\ \text{false} \end{array}$
fong-spivak-seven-sketches_0017 DIA_0017	21	—	—	—	
fong-spivak-seven-sketches_0018 DIA_0018	23	—	—	—	$\underline{3} \rightarrow \underline{3}$
fong-spivak-seven-sketches_0019 DIA_0019	23	—	—	—	surjective func $\underline{3} \twoheadrightarrow \underline{2}$
fong-spivak-seven-sketches_0020 DIA_0020	23	—	—	—	injective function $\underline{2} \hookrightarrow \underline{3}$
fong-spivak-seven-sketches_0021 DIA_0021	23	—	—	—	$\underline{2} \twoheadrightarrow \underline{3}$
fong-spivak-seven-sketches_0022 DIA_0022	23	—	—	—	bijjective funct $\underline{3} \xrightarrow{\cong} \underline{3}$
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fong-spivak-seven-sketches_1 DIA_0027	23	—	—	—	
fong-spivak-seven-sketches_1 DIA_0028	24	—	—	—	
fong-spivak-seven-sketches_1 DIA_0029	24	—	—	—	
fong-spivak-seven-sketches_1 DIA_0030	24	—	—	—	
fong-spivak-seven-sketches_1 DIA_0031	25	—	—	—	
fong-spivak-seven-sketches_1 DIA_0032	25	—	—	—	
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fong-spivak-seven-sketches_1 DIA_0034	27	—	—	—	

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fong-spivak-seven-sketches_0038	29	—	—	—	
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fong-spivak-seven-sketches_4 DIA_0045	34	—	—	—	
fong-spivak-seven-sketches_3 DIA_0046	36	—	—	—	

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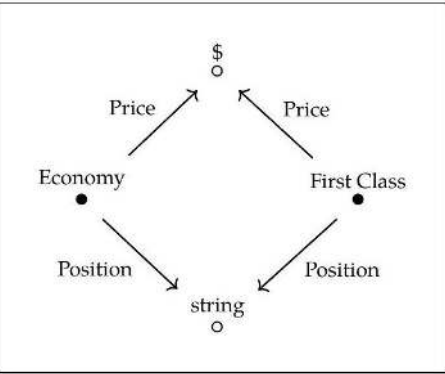
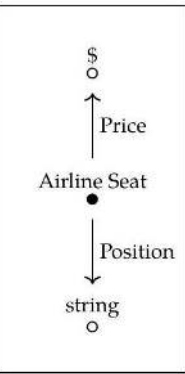
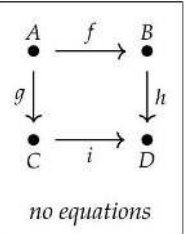
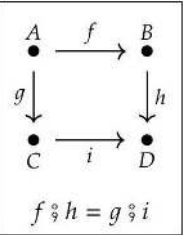
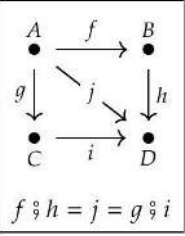
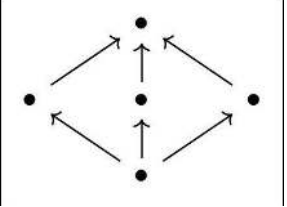
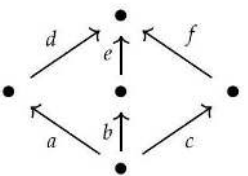
Identifier	Page	Conf.	LaTeX source	Rendered	Image
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fong-spivak-seven-sketches_0057	52	—	—	—	
fong-spivak-seven-sketches_0058	55	—	—	—	$x \leq y$
fong-spivak-seven-sketches_0059	55	—	—	—	x_0
fong-spivak-seven-sketches_0060	55	—	—	—	$x_0 \leq x_1$
fong-spivak-seven-sketches_0061	55	—	—	—	$x_0 \leq x_1 \leq x_2$
fong-spivak-seven-sketches_0062	55	—	—	—	$x_0 \leq x_1 \leq x_2 \leq x_3$
fong-spivak-seven-sketches_0063	56	—	—	—	
fong-spivak-seven-sketches_0064	56	—	—	—	
fong-spivak-seven-sketches_0065	56	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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fong-spivak-seven-sketches_0067	56	—	—	—	
fong-spivak-seven-sketches_0068	56	—	—	—	$\frac{x}{y}$
fong-spivak-seven-sketches_0069	56	—	—	—	$x \otimes y$
fong-spivak-seven-sketches_0070	57	—	—	—	$\frac{\quad}{I} \quad \text{nothing}$
fong-spivak-seven-sketches_0071	57	—	—	—	
fong-spivak-seven-sketches_0072	57	—	—	—	$\frac{3.14}{3.14} \quad \frac{\quad}{-1} \quad \left. \frac{3.14}{-1} \right\} = \frac{2.14}{2.14} \quad \frac{\quad}{0} = \text{nothing}$
fong-spivak-seven-sketches_0073	57	—	—	—	
fong-spivak-seven-sketches_0074	57	—	—	—	$\frac{\quad}{x}$
fong-spivak-seven-sketches_0075	57	—	—	—	
fong-spivak-seven-sketches_0076	58	—	—	—	
fong-spivak-seven-sketches_0077	58	—	—	—	
fong-spivak-seven-sketches_0078	58	—	—	—	$\frac{\text{nothing}}{x} \quad \text{---} \quad x$
fong-spivak-seven-sketches_0079	58	—	—	—	
fong-spivak-seven-sketches_0080	58	—	—	—	

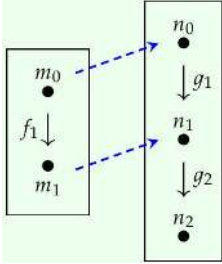
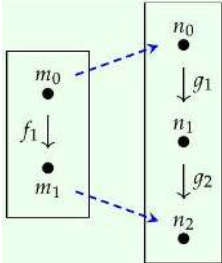
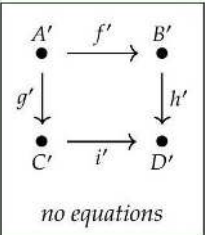
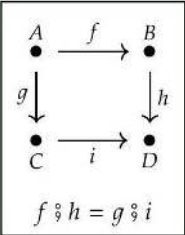
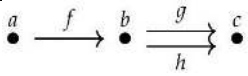
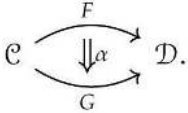
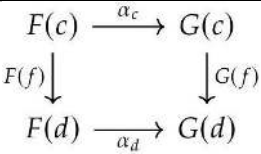
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fong-spivak-seven-sketches_61 DIA_0083	61	—	—	—	
fong-spivak-seven-sketches_62 DIA_0084	62	—	—	—	
fong-spivak-seven-sketches_62 DIA_0085	62	—	—	—	
fong-spivak-seven-sketches_63 DIA_0086	63	—	—	—	
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fong-spivak-seven-sketches_2 DIA_0094	79	—	—	—	

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fong-spivak-seven-sketches_0098	91	—	—	—	
fong-spivak-seven-sketches_0099	91	—	—	—	<pre> schema mySchema = { nodes Employee, Department; attributes DName : Department -> string, FName : Employee -> string; arrows Mngr : Employee -> Employee, WorksIn : Employee -> Department, Secr : Department -> Employee; equations Department.Secr.WorksIn = Department, Employee.Mngr.WorksIn = Employee.WorksIn; } </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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fong-spivak-seven-sketches_0101	95	—	—	—	
fong-spivak-seven-sketches_0102	96	—	—	—	$\text{Free_square} :=$ 
fong-spivak-seven-sketches_0103	96	—	—	—	$\text{Comm_square} :=$ 
fong-spivak-seven-sketches_0104	96	—	—	—	
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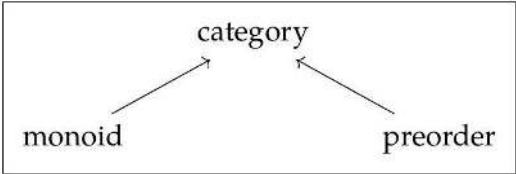
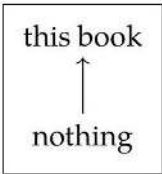
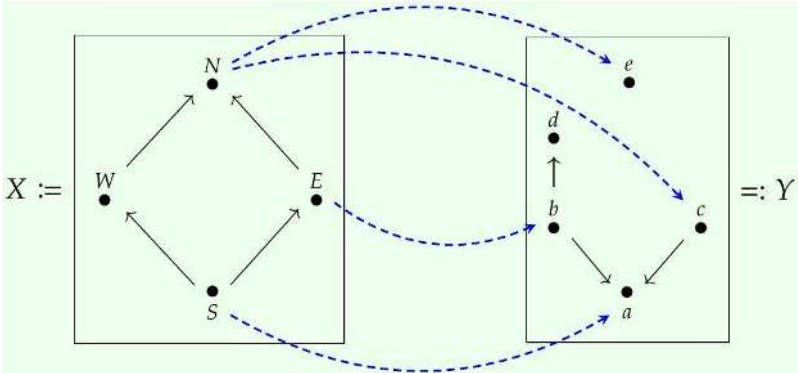
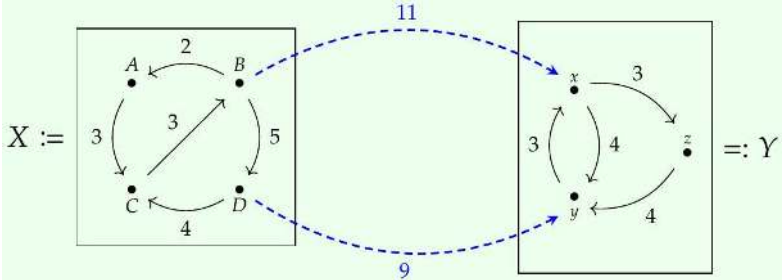
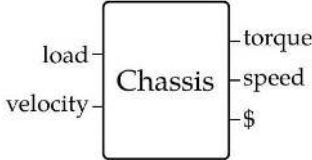
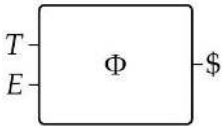
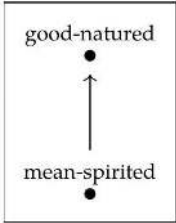
Identifier	Page	Conf.	LaTeX source	Rendered	Image
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fong-spivak-seven-sketches_0110	101	—	—	—	
fong-spivak-seven-sketches_0111	102	—	—	—	
fong-spivak-seven-sketches_0112	102	—	—	—	
fong-spivak-seven-sketches_0113	102	—	—	—	
fong-spivak-seven-sketches_0114	102	—	—	—	
fong-spivak-seven-sketches_0115	103	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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fong-spivak-seven-sketches_104 DIA_0119	104	—	—	—	Comm_square :=  $f \circ h = g \circ i$
fong-spivak-seven-sketches_106 DIA_0120	106	—	—	—	1.  $s \circ s = z$
fong-spivak-seven-sketches_106 DIA_0121	106	—	—	—	
fong-spivak-seven-sketches_106 DIA_0122	106	—	—	—	$f \circ g = f \circ h$
fong-spivak-seven-sketches_107 DIA_0123	107	—	—	—	
fong-spivak-seven-sketches_107 DIA_0124	107	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_108 DIA_0125	108	—	—	—	
fong-spivak-seven-sketches_109 DIA_0126	109	—	—	—	$\mathbf{Gr} := \begin{array}{c} \text{Arrow} \xrightarrow{\text{source}} \text{Vertex} \\ \bullet \xrightarrow{\text{target}} \bullet \\ \text{no equations} \end{array}$
fong-spivak-seven-sketches_110 DIA_0127	110	—	—	—	$\begin{array}{ccc} G(\text{Arrow}) & \xrightarrow{\alpha_{\text{Arrow}}} & H(\text{Arrow}) \\ G(\text{source}) \downarrow & & \downarrow H(\text{source}) \\ G(\text{Vertex}) & \xrightarrow{\alpha_{\text{Vertex}}} & H(\text{Vertex}) \end{array}$
fong-spivak-seven-sketches_110 DIA_0128	110	—	—	—	$\begin{array}{ccc} G(\text{Arrow}) & \xrightarrow{\alpha_{\text{Arrow}}} & H(\text{Arrow}) \\ G(\text{target}) \downarrow & & \downarrow H(\text{target}) \\ G(\text{Vertex}) & \xrightarrow{\alpha_{\text{Vertex}}} & H(\text{Vertex}) \end{array}$
fong-spivak-seven-sketches_112 DIA_0129	112	—	—	—	
fong-spivak-seven-sketches_113 DIA_0130	113	—	—	—	
fong-spivak-seven-sketches_114 DIA_0131	114	—	—	—	$\mathcal{C} \xrightarrow{F} \mathcal{D} \begin{array}{c} \xrightarrow{I} \\ \Downarrow \alpha \\ \xrightarrow{J} \end{array} \mathbf{Set} \quad \leadsto \quad \mathcal{C} \begin{array}{c} \xrightarrow{F \circ I} \\ \Downarrow \alpha_F \\ \xrightarrow{F \circ J} \end{array} \mathbf{Set}$
fong-spivak-seven-sketches_115 DIA_0132	115	—	—	—	$\begin{array}{ccc} \mathcal{C}(c, Rd) & \xrightarrow{\alpha_{c,d}} & \mathcal{D}(Lc, d) \\ \mathcal{C}(f, Rg) \downarrow & & \downarrow \mathcal{D}(Lf, g) \\ \mathcal{C}(c', Rd') & \xrightarrow{\alpha_{c',d'}} & \mathcal{D}(Lc', d') \end{array}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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fong-spivak-seven-sketches_122 DIA_0137	122	—	—	—	
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fong-spivak-seven-sketches_0142	125	—	—	—	
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Identifier	Page	Conf.	LaTeX source	Rendered	Image																									
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fong-spivak-seven-sketches_144 DIA_0160	144	—	—	—																										
fong-spivak-seven-sketches_144 DIA_0161	144	—	—	—	<table><thead><tr><th>Col(Φ)</th><th>A</th><th>B</th><th>x</th><th>y</th></tr></thead><tbody><tr><td>A</td><td>0</td><td>2</td><td>5</td><td>8</td></tr><tr><td>B</td><td>∞</td><td>0</td><td>∞</td><td>∞</td></tr><tr><td>x</td><td>0</td><td>0</td><td>0</td><td>3</td></tr><tr><td>y</td><td>0</td><td>0</td><td>4</td><td>0</td></tr></tbody></table>	Col(Φ)	A	B	x	y	A	0	2	5	8	B	∞	0	∞	∞	x	0	0	0	3	y	0	0	4	0
Col(Φ)	A	B	x	y																										
A	0	2	5	8																										
B	∞	0	∞	∞																										
x	0	0	0	3																										
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fong-spivak-seven-sketches DIA_0176	152	—	—	—	<pre> graph LR f1 --> f2 f2 --> f3 f3 --> f1 f2 --> f4 f4 --> f5 f5 --> f3 f4 --> f2 f5 --> f4 </pre>
fong-spivak-seven-sketches DIA_0177	152	—	—	—	<pre> graph LR pain --> P1[Person 1] P1 -- sound --> P2[Person 2] P2 -- fury --> P1 P2 -- complaint --> Out </pre>
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fong-spivak-seven-sketches DIA_0179	152	—	—	—	<pre> graph LR pain --> P1[Person 1] P1 -- sound --> P1 P1 -- fury --> P2[Person 2] P2 -- complaint --> Out </pre>
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Identifier	Page	Conf.	LaTeX source	Rendered	Image
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fong-spivak-seven-sketches_0183 DIA_0183	153	—	—	—	$ \epsilon_c $
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fong-spivak-seven-sketches_0185 DIA_0185	153	—	—	—	$ \eta_c \otimes c^* \quad c^* \otimes \epsilon_c $
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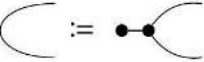
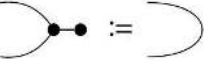
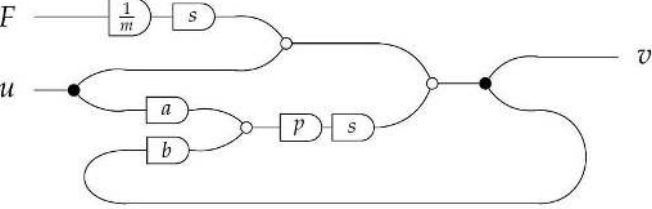
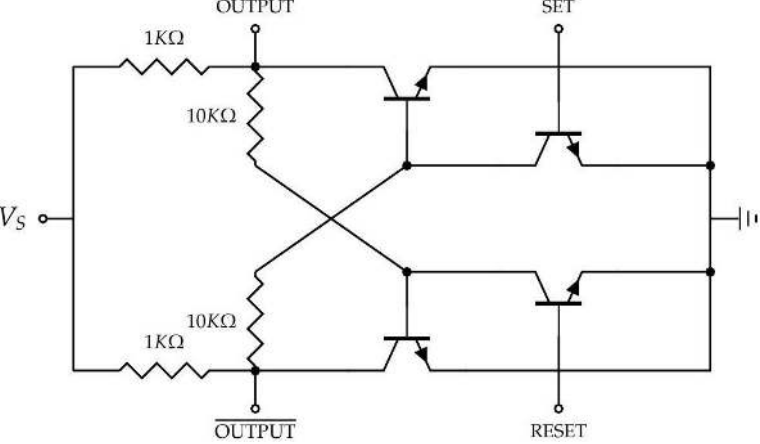
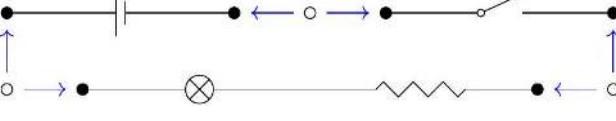
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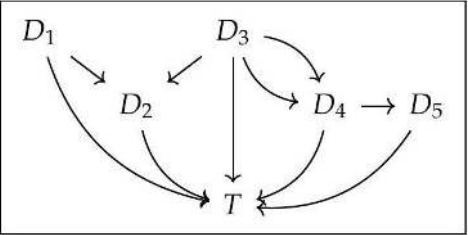
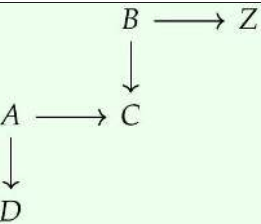
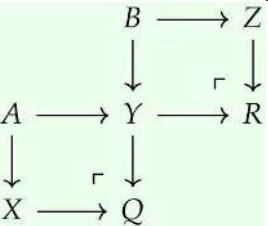
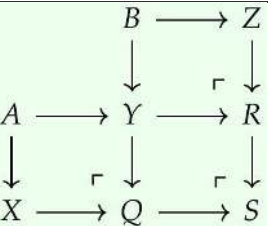
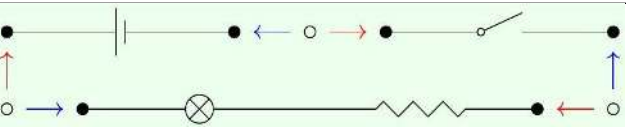
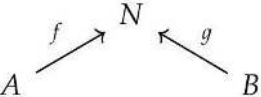
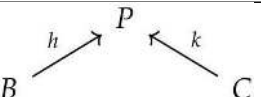
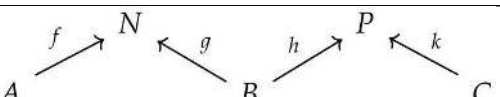
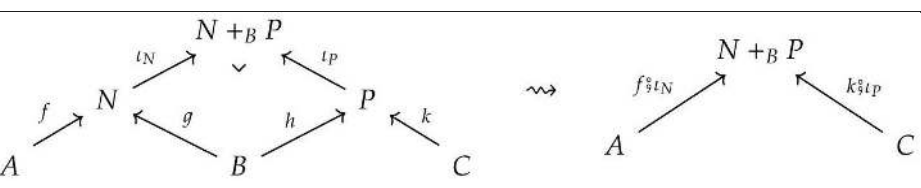
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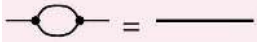
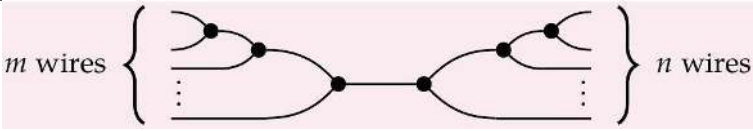
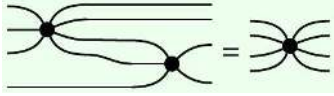
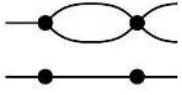
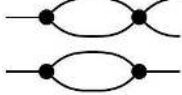
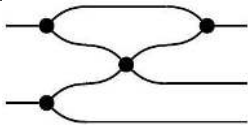
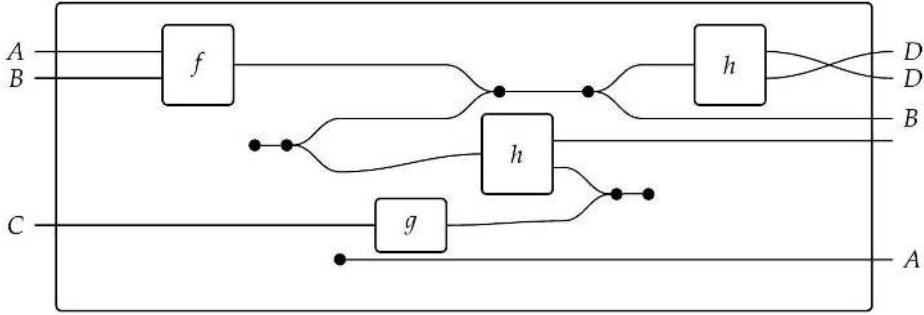
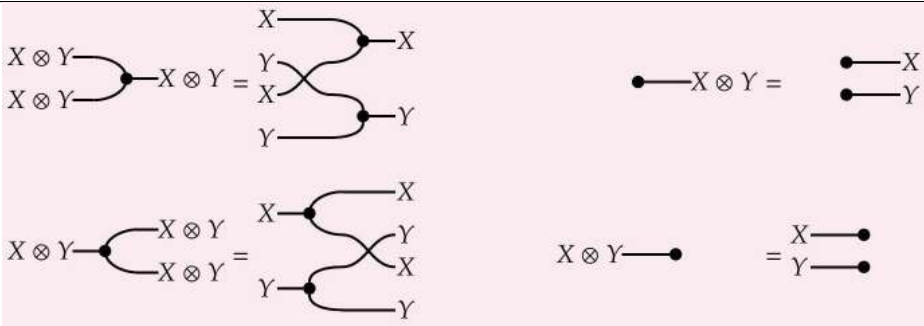
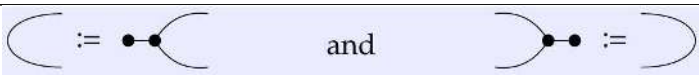
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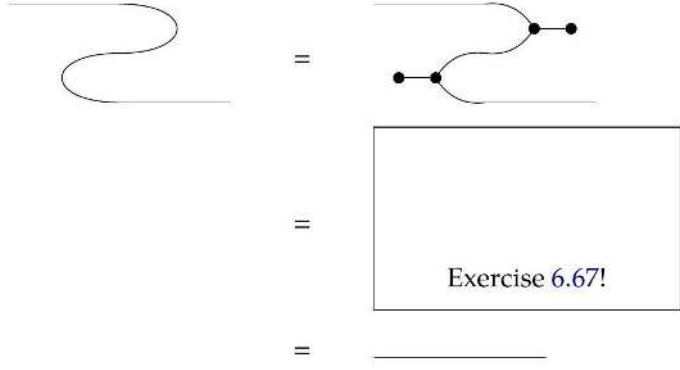
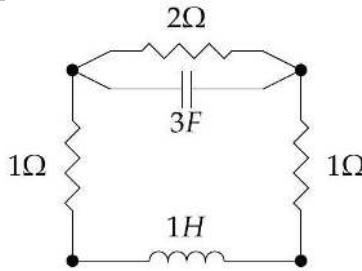
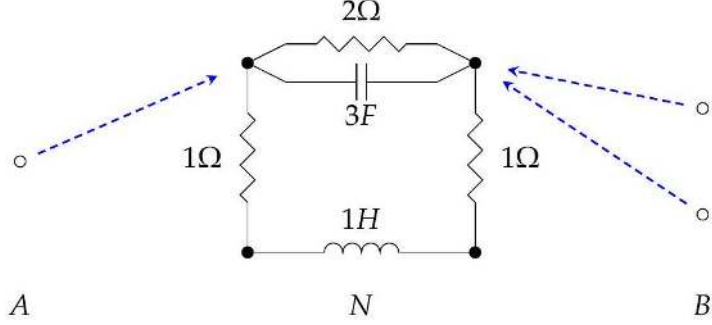
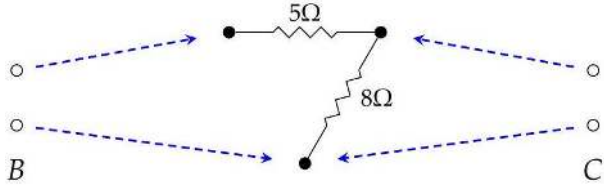
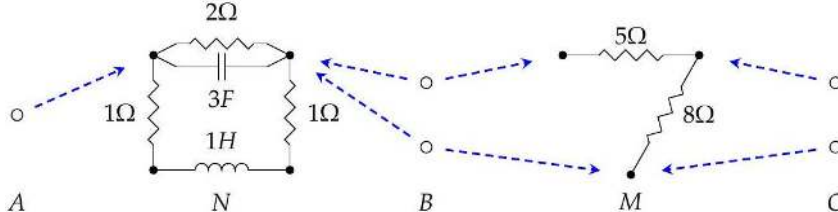
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fong-spivak-seven-sketches_1 DIA_0297	216	—	—	—	$ \begin{array}{ccc} P(S) \times P(T) & \xrightarrow{\varphi_{S,T}} & P(S \times T) \\ \text{im}_f \times \text{im}_g \downarrow & & \downarrow \text{im}_f \times g \\ P(S') \times P(T') & \xrightarrow{\varphi_{S',T'}} & P(S' \times T') \end{array} $
fong-spivak-seven-sketches_1 DIA_0298	217	—	—	—	
fong-spivak-seven-sketches_1 DIA_0299	217	—	—	—	
fong-spivak-seven-sketches_1 DIA_0300	217	—	—	—	
fong-spivak-seven-sketches_1 DIA_0301	218	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_1 DIA_0302	218	—	—	—	
fong-spivak-seven-sketches_1 DIA_0303	219	—	—	—	
fong-spivak-seven-sketches_1 DIA_0304	220	—	—	—	
fong-spivak-seven-sketches_1 DIA_0305	220	—	—	—	
fong-spivak-seven-sketches_1 DIA_0306	220	—	—	—	
fong-spivak-seven-sketches_1 DIA_0307	220	—	—	—	
fong-spivak-seven-sketches_1 DIA_0308	220	—	—	—	
fong-spivak-seven-sketches_1 DIA_0309	220	—	—	—	$b := \bullet \text{---} \text{---} \bullet$ and $s := \bullet \text{---} \diagup \text{---} \bullet$
fong-spivak-seven-sketches_1 DIA_0310	221	—	—	—	
fong-spivak-seven-sketches_1 DIA_0311	221	—	—	—	
fong-spivak-seven-sketches_1 DIA_0312	221	—	—	—	
fong-spivak-seven-sketches_1 DIA_0313	221	—	—	—	
fong-spivak-seven-sketches_1 DIA_0314	221	—	—	—	

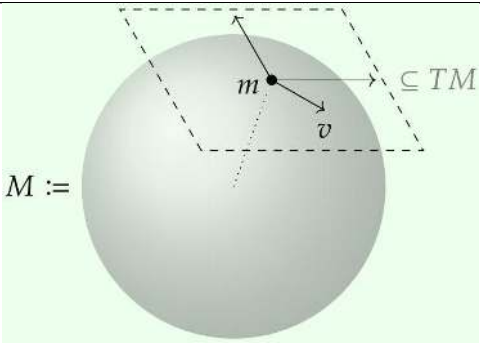
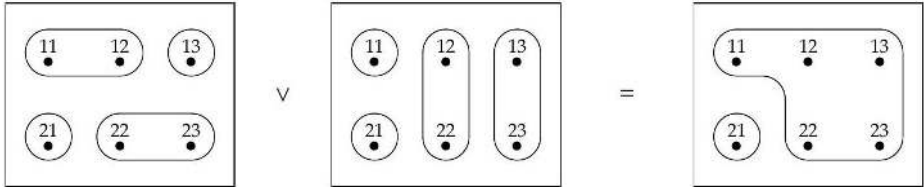
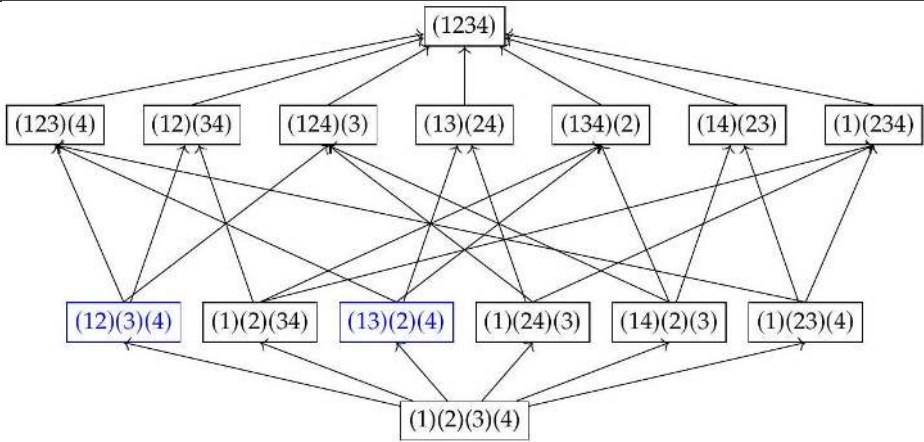
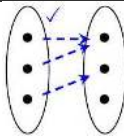
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fong-spivak-seven-sketches_0317	222	—	—	—	
fong-spivak-seven-sketches_0318	222	—	—	—	
fong-spivak-seven-sketches_0319	222	—	—	—	
fong-spivak-seven-sketches_0320	222	—	—	—	
fong-spivak-seven-sketches_0321	223	—	—	—	
fong-spivak-seven-sketches_0322	223	—	—	—	
fong-spivak-seven-sketches_0323	223	—	—	—	$\eta := \emptyset$
fong-spivak-seven-sketches_0324	223	—	—	—	$\emptyset =: \epsilon$
fong-spivak-seven-sketches_0325	224	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_ DIA_0326	224	—	—	—	
fong-spivak-seven-sketches_ DIA_0327	226	—	—	—	
fong-spivak-seven-sketches_ DIA_0328	227	—	—	—	
fong-spivak-seven-sketches_ DIA_0329	228	—	—	—	
fong-spivak-seven-sketches_ DIA_0330	228	—	—	—	
fong-spivak-seven-sketches_ DIA_0331	229	—	—	—	$\varphi :=$
fong-spivak-seven-sketches_ DIA_0332	229	—	—	—	
fong-spivak-seven-sketches_ DIA_0333	230	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_234 DIA_0334	234	—	—	—	
fong-spivak-seven-sketches_234 DIA_0335	234	—	—	—	
fong-spivak-seven-sketches_236 DIA_0336	236	—	—	—	
fong-spivak-seven-sketches_237 DIA_0337	237	—	—	—	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & & \downarrow & \lrcorner & \downarrow \\ D & \longrightarrow & E & \longrightarrow & F \end{array}$
fong-spivak-seven-sketches_237 DIA_0338	237	—	—	—	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & \lrcorner & \downarrow & & \downarrow \\ D & \longrightarrow & E & \longrightarrow & F \end{array}$
fong-spivak-seven-sketches_237 DIA_0339	237	—	—	—	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \\ A' & \longrightarrow & B' & \longrightarrow & C' \end{array}$
fong-spivak-seven-sketches_238 DIA_0340	238	—	—	—	$\begin{array}{ccc} A & \xrightarrow{\text{id}_A} & A \\ \text{id}_A \downarrow & \lrcorner & \downarrow f \\ A & \xrightarrow{f} & B \end{array} \quad \begin{array}{ccc} A & \xrightarrow{f} & B \\ f \downarrow & \lrcorner & \downarrow \text{id}_B \\ B & \xrightarrow{\text{id}_B} & B \end{array}$
fong-spivak-seven-sketches_238 DIA_0341	238	—	—	—	$\begin{array}{ccc} A' & \xrightarrow{f'} & B' \\ i' \downarrow \cong & \lrcorner & \cong \downarrow i \\ A & \xrightarrow{f} & B \end{array}$
fong-spivak-seven-sketches_238 DIA_0342	238	—	—	—	$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \parallel & \lrcorner & \parallel \\ A & \xrightarrow{f} & B \end{array}$
fong-spivak-seven-sketches_238 DIA_0343	238	—	—	—	$\begin{array}{ccc} A' & \xrightarrow{g} & A \\ f' \downarrow & \lrcorner & \downarrow f \\ B' & \xrightarrow{h} & B \end{array}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_1 DIA_0344	239	—	—	—	
fong-spivak-seven-sketches_1 DIA_0345	240	—	—	—	$\begin{array}{ccc} X & \xrightarrow{!} & 1 \\ \downarrow m & \lrcorner & \downarrow \text{true} \\ Y & \xrightarrow{\ulcorner m \urcorner} & \Omega \end{array}$
fong-spivak-seven-sketches_1 DIA_0346	240	—	—	—	$\begin{array}{ccc} \{Y \mid p\} & \xrightarrow{!} & 1 \\ \downarrow & \lrcorner & \downarrow \text{true} \\ Y & \xrightarrow{p} & \Omega \end{array}$
fong-spivak-seven-sketches_1 DIA_0347	245	—	—	—	$\begin{array}{ccc} & \xrightarrow{P} & \\ \mathcal{C}^{\text{op}} & \Downarrow \alpha & \mathbf{Set}. \\ & \xrightarrow{Q} & \end{array}$
fong-spivak-seven-sketches_1 DIA_0348	245	—	—	—	ArShp := $\begin{array}{ccc} \text{Vertex} & \xrightarrow{\text{src}} & \text{Pure Arrow} \\ \bullet & \xrightarrow{\text{tgt}} & \bullet \end{array}$
fong-spivak-seven-sketches_1 DIA_0349	245	—	—	—	Vertex := Pure Arrow := Diagram showing a box labeled 'Vertex' with a dot inside, and a box labeled 'Pure Arrow' with an arrow inside. Two curved arrows, labeled 'src' and 'tgt', connect the dot in the 'Vertex' box to the arrow in the 'Pure Arrow' box.
fong-spivak-seven-sketches_1 DIA_0350	245	—	—	—	$P := \begin{array}{ccc} \bullet & \xleftarrow{\quad} \bullet & \xrightarrow{\quad} \bullet \\ & \Downarrow & \\ \bullet & \xleftarrow{\quad} \bullet & \xrightarrow{\quad} \bullet \end{array} \quad \begin{array}{c} \bullet \\ \curvearrowright \\ \bullet \end{array} \quad \begin{array}{c} \bullet \\ \xrightarrow{\quad} \bullet \end{array} : \mathbf{ArShp}^{\text{op}} \rightarrow \mathbf{Set}$
fong-spivak-seven-sketches_1 DIA_0351	247	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_0352	250	—	—	—	$X :=$ $Y :=$ $f \downarrow$
fong-spivak-seven-sketches_0353	251	—	—	—	s_1 s_2 s_3 s_4 s_5 s_6
fong-spivak-seven-sketches_0354	251	—	—	—	$s_1 _V = s_2 _V = s_3 _V$ $s_4 _V = s_5 _V = s_6 _V$
fong-spivak-seven-sketches_0355	252	—	—	—	g_1 g_2
fong-spivak-seven-sketches_0356	252	—	—	—	glued section

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_0357	253	—	—	—	
fong-spivak-seven-sketches_0358	261	—	—	—	$\begin{array}{ccc} \forall_t p & \longrightarrow & 1 \\ \downarrow & \lrcorner & \downarrow \text{true}^T \\ S & \xrightarrow{p'} & \Omega^T \end{array}$
fong-spivak-seven-sketches_0359	262	—	—	—	$\begin{array}{ccc} \{S \times T \mid p\} & \xrightarrow{\quad} & S \times T \\ \downarrow & & \downarrow \pi_S \\ \exists_t p & \xrightarrow{\quad} & S \end{array}$
fong-spivak-seven-sketches_0360	271	—	—	—	
fong-spivak-seven-sketches_0361	271	—	—	—	$\begin{array}{c} (12) = \bullet \quad * \\ \uparrow \\ (1)(2) = \bullet \quad * \end{array}$
fong-spivak-seven-sketches_0362	272	—	—	—	
fong-spivak-seven-sketches_0363	273	—	—	—	

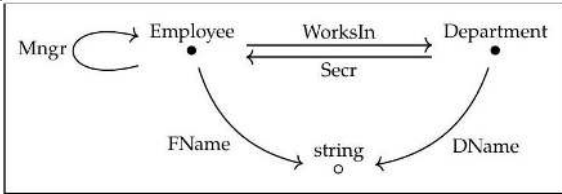
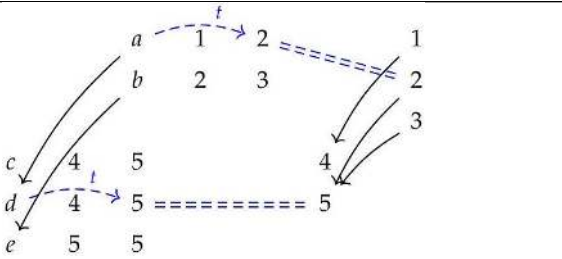
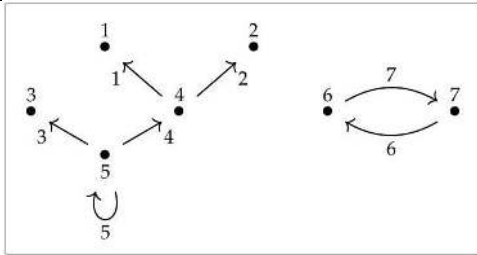
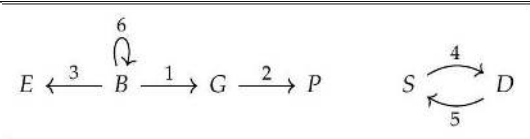
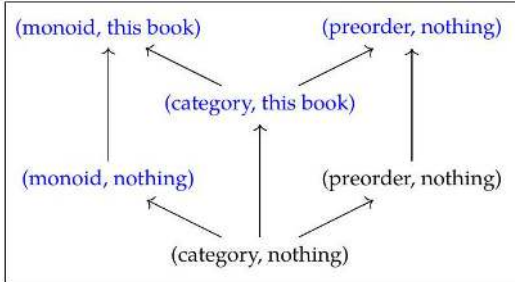
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fong-spivak-seven-sketches_273 DIA_0365	273	—	—	—	
fong-spivak-seven-sketches_273 DIA_0366	273	—	—	—	
fong-spivak-seven-sketches_274 DIA_0367	274	—	—	—	$\bullet \mapsto p_1$ $* \mapsto p_2$ $\circ \mapsto p_3$
fong-spivak-seven-sketches_275 DIA_0368	275	—	—	—	
fong-spivak-seven-sketches_275 DIA_0369	275	—	—	—	
fong-spivak-seven-sketches_275 DIA_0370	275	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_276 DIA_0371	276	—	—	—	
fong-spivak-seven-sketches_276 DIA_0372	276	—	—	—	
fong-spivak-seven-sketches_276 DIA_0373	276	—	—	—	
fong-spivak-seven-sketches_278 DIA_0374	278	—	—	—	
fong-spivak-seven-sketches_278 DIA_0375	278	—	—	—	
fong-spivak-seven-sketches_278 DIA_0376	278	—	—	—	
fong-spivak-seven-sketches_278 DIA_0377	278	—	—	—	
fong-spivak-seven-sketches_278 DIA_0378	278	—	—	—	
fong-spivak-seven-sketches_278 DIA_0379	278	—	—	—	
fong-spivak-seven-sketches_279 DIA_0380	279	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_279 DIA_0381	279	—	—	—	
fong-spivak-seven-sketches_279 DIA_0382	279	—	—	—	
fong-spivak-seven-sketches_279 DIA_0383	279	—	—	—	
fong-spivak-seven-sketches_279 DIA_0384	279	—	—	—	
fong-spivak-seven-sketches_279 DIA_0385	279	—	—	—	$c =$
fong-spivak-seven-sketches_279 DIA_0386	279	—	—	—	$g!(c) =$
fong-spivak-seven-sketches_279 DIA_0387	279	—	—	—	$d =$
fong-spivak-seven-sketches_279 DIA_0388	279	—	—	—	$e =$
fong-spivak-seven-sketches_279 DIA_0389	279	—	—	—	$g^*(d) =$
fong-spivak-seven-sketches_279 DIA_0390	279	—	—	—	$g^*(e) =$
fong-spivak-seven-sketches_280 DIA_0391	280	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_0392	280	—	—	—	
fong-spivak-seven-sketches_0393	281	—	—	—	
fong-spivak-seven-sketches_0394	285	—	—	—	
fong-spivak-seven-sketches_0395	285	—	—	—	
fong-spivak-seven-sketches_0396	285	—	—	—	
fong-spivak-seven-sketches_0397	290	—	—	—	
fong-spivak-seven-sketches_0398	290	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_0399	290	—	—	—	$G_1 = \begin{array}{ccc} \bullet & \xrightarrow{f} & \bullet \\ \bullet & \xrightarrow{g} & \bullet \\ f = g \end{array} \quad G_2 = \begin{array}{c} f \\ \downarrow \\ \bullet \\ a \\ f = a \end{array} \quad G_3 = \begin{array}{ccc} \bullet & \xrightarrow{f} & \bullet \\ g \downarrow & & \downarrow h \\ \bullet & \xrightarrow{i} & \bullet \\ f \circ h = g \circ i \end{array} \quad G_4 = \begin{array}{ccc} \bullet & \xrightarrow{f} & \bullet \\ g \downarrow & & \downarrow h \\ \bullet & & \bullet \\ \text{no equations} \end{array}$
fong-spivak-seven-sketches_0400	291	—	—	—	
fong-spivak-seven-sketches_0401	291	—	—	—	
fong-spivak-seven-sketches_0402	291	—	—	—	
fong-spivak-seven-sketches_0403	292	—	—	—	$\begin{array}{c} s \\ \downarrow \\ \bullet \\ z \\ s \circ s = z \end{array}$
fong-spivak-seven-sketches_0404	292	—	—	—	$\begin{array}{ccccc} a & \xrightarrow{f} & b & \xrightarrow[g]{h} & c \\ f \circ g = f \circ h \end{array}$
fong-spivak-seven-sketches_0405	292	—	—	—	
fong-spivak-seven-sketches_0406	293	—	—	—	$\begin{array}{ccccc} F(c) & \xrightarrow{\alpha_c} & G(c) & \xrightarrow{\beta_c} & H(c) \\ F(f) \downarrow & & G(f) \downarrow & & H(f) \downarrow \\ F(c') & \xrightarrow{\alpha_c} & G(c') & \xrightarrow{\beta_c} & H(c') \end{array}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image																
fong-spivak-seven-sketches_1 DIA_0407	293	—	—	—	$\begin{array}{ccc} F(c) & \xrightarrow{\text{id}_{F(c)}} & F(c) \\ F(f) \downarrow & & \downarrow F(f) \\ F(c') & \xrightarrow{\text{id}_{F(c')}} & F(c') \end{array}$																
fong-spivak-seven-sketches_1 DIA_0408	294	—	—	—																	
fong-spivak-seven-sketches_1 DIA_0409	294	—	—	—																	
fong-spivak-seven-sketches_1 DIA_0410	294	—	—	—	Arrow source target Vertex Gr $\xrightarrow{G}$ State next DDS $\xrightarrow{\hspace{1cm}}$ Set <table><tr><th>State</th><th>next</th></tr><tr><td>1</td><td>4</td></tr><tr><td>2</td><td>4</td></tr><tr><td>3</td><td>5</td></tr><tr><td>4</td><td>5</td></tr><tr><td>5</td><td>5</td></tr><tr><td>6</td><td>7</td></tr><tr><td>7</td><td>6</td></tr></table>	State	next	1	4	2	4	3	5	4	5	5	5	6	7	7	6
State	next																				
1	4																				
2	4																				
3	5																				
4	5																				
5	5																				
6	7																				
7	6																				
fong-spivak-seven-sketches_1 DIA_0411	295	—	—	—																	
fong-spivak-seven-sketches_1 DIA_0412	295	—	—	—																	
fong-spivak-seven-sketches_1 DIA_0413	298	—	—	—																	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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fong-spivak-seven-sketches_DIA_0416	302	—	—	—	
fong-spivak-seven-sketches_DIA_0417	302	—	—	—	
fong-spivak-seven-sketches_DIA_0418	303	—	—	—	
fong-spivak-seven-sketches_DIA_0419	303	—	—	—	
fong-spivak-seven-sketches_DIA_0420	305	—	—	—	

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fong-spivak-seven-sketches_305 DIA_0423	305	—	—	—	
fong-spivak-seven-sketches_306 DIA_0424	306	—	—	—	
fong-spivak-seven-sketches_306 DIA_0425	306	—	—	—	
fong-spivak-seven-sketches_307 DIA_0426	307	—	—	—	
fong-spivak-seven-sketches_307 DIA_0427	307	—	—	—	
fong-spivak-seven-sketches_307 DIA_0428	307	—	—	—	
fong-spivak-seven-sketches_309 DIA_0429	309	—	—	—	
fong-spivak-seven-sketches_309 DIA_0430	309	—	—	—	
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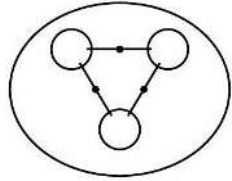
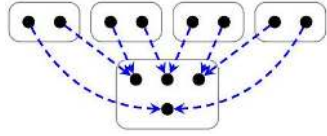
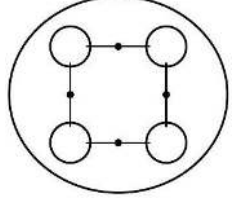
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fong-spivak-seven-sketches DIA_0433	310	—	—	—	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$
fong-spivak-seven-sketches DIA_0434	310	—	—	—	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$
fong-spivak-seven-sketches DIA_0435	310	—	—	—	$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$
fong-spivak-seven-sketches DIA_0436	310	—	—	—	
fong-spivak-seven-sketches DIA_0437	311	—	—	—	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$
fong-spivak-seven-sketches DIA_0438	311	—	—	—	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$
fong-spivak-seven-sketches DIA_0439	311	—	—	—	
fong-spivak-seven-sketches DIA_0440	311	—	—	—	
fong-spivak-seven-sketches DIA_0441	311	—	—	—	
fong-spivak-seven-sketches DIA_0442	311	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches_311 DIA_0443	311	—	—	—	
fong-spivak-seven-sketches_311 DIA_0444	311	—	—	—	
fong-spivak-seven-sketches_312 DIA_0445	312	—	—	—	
fong-spivak-seven-sketches_312 DIA_0446	312	—	—	—	
fong-spivak-seven-sketches_312 DIA_0447	312	—	—	—	
fong-spivak-seven-sketches_312 DIA_0448	312	—	—	—	
fong-spivak-seven-sketches_312 DIA_0449	312	—	—	—	
fong-spivak-seven-sketches_312 DIA_0450	312	—	—	—	
fong-spivak-seven-sketches_312 DIA_0451	312	—	—	—	
fong-spivak-seven-sketches_312 DIA_0452	312	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches DIA_0453	316	—	<pre> \begin{lstlisting}[mathescape=true] \([f, g]: A \sqcup B \longrightarrow T\) apple1 \mapsto \(\a\) banana1 \mapsto \(\b\) pear1 \mapsto \(\p\) cherry1 \mapsto \(\c\) orange1 \mapsto \(\o\) apple2 \mapsto \(\e\) tomato2 \mapsto \(\o\) mango2 \mapsto \(\o\). \end{lstlisting} </pre>	(not rendered)	$[f, g]: A \sqcup B \longrightarrow T$ $\text{apple1} \mapsto a$ $\text{banana1} \mapsto b$ $\text{pear1} \mapsto p$ $\text{cherry1} \mapsto c$ $\text{orange1} \mapsto o$ $\text{apple2} \mapsto e$ $\text{tomato2} \mapsto o$ $\text{mango2} \mapsto o.$
fong-spivak-seven-sketches DIA_0454	316	—	—	—	A commutative diagram with nodes A, B, C, and D. Arrows: $A \xrightarrow{\iota_A} A+B \xleftarrow{\iota_B} B$, $A \xrightarrow{f} C$, $B \xrightarrow{g} C$, $C \xrightarrow{h} D$. A central arrow $[f, g]: A+B \rightarrow C$ is shown.
fong-spivak-seven-sketches DIA_0455	316	—	—	—	A commutative diagram with nodes A, B, and A+B. Arrows: $A \xrightarrow{\iota_A} A+B \xleftarrow{\iota_B} B$, $A \xrightarrow{\iota_A} A+B$, $B \xrightarrow{\iota_B} A+B$. A central arrow $\text{id}_{A+B}: A+B \rightarrow A+B$ is shown.
fong-spivak-seven-sketches DIA_0456	317	—	—	—	A commutative diagram with nodes A, B, C, D, E, and F. Arrows: $A \xrightarrow{\iota_A} A+B \xleftarrow{\iota_B} B$, $A \xrightarrow{f} C$, $B \xrightarrow{g} D$, $C \xrightarrow{\iota_C} C+D \xleftarrow{\iota_D} D$, $C+D \xrightarrow{h+k} E+F$, $C \xrightarrow{h \circ \iota_C} E+F$, $D \xrightarrow{k \circ \iota_D} E+F$.
fong-spivak-seven-sketches DIA_0457	318	—	—	—	A complex diagram showing a pushout of two maps f and g from a set of points to another set of points. The diagram is labeled "pushout".

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fong-spivak-seven-sketches_DIA_0458	318	—	—	—	
fong-spivak-seven-sketches_DIA_0459	318	—	—	—	
fong-spivak-seven-sketches_DIA_0460	318	—	—	—	
fong-spivak-seven-sketches_DIA_0461	318	—	—	—	
fong-spivak-seven-sketches_DIA_0462	319	—	—	—	
fong-spivak-seven-sketches_DIA_0463	319	—	—	—	
fong-spivak-seven-sketches_DIA_0464	320	—	—	—	
fong-spivak-seven-sketches_DIA_0465	320	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches DIA_0466	320	—	—	—	$ \begin{array}{ccc} X + X & \xrightarrow{[\text{id}, \text{id}]} & X \\ [\text{id}, \text{id}] \downarrow & & \downarrow \text{id} \\ X & \xrightarrow{\text{id}} & X \end{array} $
fong-spivak-seven-sketches DIA_0467	320	—	—	—	$ \begin{array}{ccc} X + X & \xrightarrow{[\text{id}, \text{id}]} & X \\ [\text{id}, \text{id}] \downarrow & & \downarrow f \\ X & \xrightarrow{g} & T \end{array} $
fong-spivak-seven-sketches DIA_0468	320	—	—	—	$ \begin{array}{ccc} X & \xrightarrow{[\text{id}, \text{id}]} & X \\ [\text{id}, \text{id}] \downarrow & & \downarrow \text{id} \\ X & \xrightarrow{\text{id}} & X \end{array} \begin{array}{c} \downarrow f \\ T \end{array} $ $g = f$
fong-spivak-seven-sketches DIA_0469	320	—	—	—	
fong-spivak-seven-sketches DIA_0470	321	—	—	—	
fong-spivak-seven-sketches DIA_0471	321	—	—	—	
fong-spivak-seven-sketches DIA_0472	322	—	—	—	$ \eta \circ x = \emptyset $ $ = \emptyset $
fong-spivak-seven-sketches DIA_0473	322	—	—	—	$ \eta \circ x \circ \epsilon = \emptyset $ $ = \emptyset $
fong-spivak-seven-sketches DIA_0474	322	—	—	—	inner circles' ports links outer circle's ports
fong-spivak-seven-sketches DIA_0475	323	—	—	—	

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fong-spivak-seven-sketches DIA_0476	323	—	—	—	
fong-spivak-seven-sketches DIA_0477	323	—	—	—	
fong-spivak-seven-sketches DIA_0478	323	—	—	—	
fong-spivak-seven-sketches DIA_0479	324	—	—	—	$ \begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C \\ \downarrow h_1 & & \downarrow h_2 & \lrcorner & \downarrow h_3 \\ A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' \end{array} $
fong-spivak-seven-sketches DIA_0480	324	—	—	—	$ \begin{array}{c} X \xrightarrow{p} C \\ \downarrow q \quad \downarrow h_1 \quad \downarrow h_2 \quad \lrcorner \quad \downarrow h_3 \\ A \xrightarrow{f} B \xrightarrow{g} C \\ \downarrow h_1 \quad \downarrow h_2 \quad \lrcorner \quad \downarrow h_3 \\ A' \xrightarrow{f'} B' \xrightarrow{g'} C' \end{array} $
fong-spivak-seven-sketches DIA_0481	324	—	—	—	$ \begin{array}{c} X \xrightarrow{p} C \\ \downarrow q \quad \downarrow h_1 \quad \downarrow h_2 \quad \lrcorner \quad \downarrow h_3 \\ A \xrightarrow{f} B \xrightarrow{g} C \\ \downarrow h_1 \quad \downarrow h_2 \quad \lrcorner \quad \downarrow h_3 \\ A' \xrightarrow{f'} B' \xrightarrow{g'} C' \end{array} $
fong-spivak-seven-sketches DIA_0482	324	—	—	—	$ \begin{array}{c} X \xrightarrow{p} C \\ \downarrow q \quad \downarrow h_1 \quad \downarrow h_2 \quad \lrcorner \quad \downarrow h_3 \\ A \xrightarrow{f} B \xrightarrow{g} C \\ \downarrow h_1 \quad \downarrow h_2 \quad \lrcorner \quad \downarrow h_3 \\ A' \xrightarrow{f'} B' \xrightarrow{g'} C' \end{array} $
fong-spivak-seven-sketches DIA_0483	324	—	—	—	$ \begin{array}{c} X \xrightarrow{p} C \\ \downarrow q \quad \downarrow h_1 \quad \downarrow h_2 \quad \lrcorner \quad \downarrow h_3 \\ A \xrightarrow{f} B \xrightarrow{g} C \\ \downarrow h_1 \quad \downarrow h_2 \quad \lrcorner \quad \downarrow h_3 \\ A' \xrightarrow{f'} B' \xrightarrow{g'} C' \end{array} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-seven-sketches DIA_0484	324	—	—	—	$ \begin{array}{ccccc} X & & & & \\ & \searrow^r & & & \\ & A \xrightarrow{f} B \xrightarrow{g} C & & & \\ & \downarrow h_1 \quad \downarrow h_2 \quad \downarrow h_3 & & & \\ & A' \xrightarrow{f'} B' \xrightarrow{g'} C' & & & \\ & \swarrow q & & & \end{array} $
fong-spivak-seven-sketches DIA_0485	325	—	—	—	$ \begin{array}{ccccc} X & & & & \\ & \searrow^r & & \searrow^p & \\ & A \xrightarrow{f} B \xrightarrow{g} C & & & \\ & \downarrow h_1 \quad \downarrow h_2 \quad \downarrow h_3 & & & \\ & A' \xrightarrow{f'} B' \xrightarrow{g'} C' & & & \\ & \swarrow q & & & \end{array} $
fong-spivak-seven-sketches DIA_0486	325	—	—	—	$ \begin{array}{ccccc} X & & & & \\ & \searrow^{g_2} & & & \\ & A \xrightarrow{\text{id}_A} A & & & \\ & \downarrow \text{id}_A \quad \downarrow f & & & \\ & A \xrightarrow{f} B & & & \\ & \swarrow g_1 & & & \end{array} $
fong-spivak-seven-sketches DIA_0487	325	—	—	—	$ \begin{array}{ccc} A' & \xrightarrow{f'} & B' \\ i' \downarrow & \lrcorner & \cong \downarrow i \\ A & \xrightarrow{f} & B \end{array} $
fong-spivak-seven-sketches DIA_0488	325	—	—	—	$ \begin{array}{ccc} A & \xrightarrow{f} & B \\ \parallel \lrcorner & & \parallel \\ A & \xrightarrow{f} & B \end{array} $
fong-spivak-seven-sketches DIA_0489	326	—	—	—	$ \begin{array}{ccccc} & & A & & \\ & & \parallel & & \\ A' & \xrightarrow{g} & A & & \\ \downarrow f' & \lrcorner & \parallel & \lrcorner & \downarrow f \\ A' & \xrightarrow{f'} & A & & \\ \downarrow f' & \searrow & \downarrow f & & \\ B' & \xrightarrow{h} & B & & \end{array} $
fong-spivak-seven-sketches DIA_0490	326	—	—	—	$ \begin{array}{ccccc} A' & \xrightarrow{g} & A & & \\ \parallel & \lrcorner & \parallel & \lrcorner & \parallel \\ A' & \xrightarrow{g} & A & & \\ \downarrow f' & \lrcorner & \parallel & \lrcorner & \downarrow f \\ A' & \xrightarrow{f'} & A & & \\ \downarrow f' & \searrow & \downarrow f & & \\ B' & \xrightarrow{h} & B & & \end{array} $
fong-spivak-seven-sketches DIA_0491	326	—	—	—	

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fong-spivak-seven-sketches DIA_0492	326	—	—	—	$ \begin{array}{ccccccc} (0,0) & \leftarrow & (0,1) & \leftarrow & (0,2) & \leftarrow & (0,3) \cdots \\ \uparrow & & \uparrow & & \uparrow & & \\ (1,0) & \leftarrow & (1,1) & \leftarrow & (1,2) & \cdots & \\ \uparrow & & \uparrow & & & & \\ (2,0) & \leftarrow & (1,1) & \cdots & & & \\ \uparrow & & & & & & \\ (3,0) & \cdots & & & & & \\ \vdots & & & & & & \end{array} $
fong-spivak-seven-sketches DIA_0493	328	—	—	—	
fong-spivak-seven-sketches DIA_0494	329	—	—	—	$ \begin{array}{c} X := \begin{array}{cccccc} & b_3 & & & & \\ a_2 & & b_2 & & & e_2 \\ a_1 & & b_1 & & c_1 & e_1 \end{array} \\ \downarrow f \\ Y := \begin{array}{ccccc} a & b & c & d & e \end{array} \end{array} $
fong-spivak-seven-sketches DIA_0495	329	—	—	—	$ \begin{array}{c} X := \begin{array}{cccccc} a_2 & & c_1 & & & e_2 \\ e_1 & & b_1 & & b_2 & b_3 & a_1 \end{array} \\ \downarrow f \\ Y := \begin{array}{ccccc} a & b & c & d & e \end{array} \end{array} $
fong-spivak-seven-sketches DIA_0496	329	—	—	—	
fong-spivak-seven-sketches DIA_0497	330	—	—	—	$ \begin{array}{ccccccccc} (a_1, b_1, c_1) & & (a_1, b_2, c_1) & & (a_1, b_3, c_1) & & (a_2, b_1, c_1) & & (a_2, b_2, c_1) & & (a_2, b_3, c_1) \\ & \swarrow & \downarrow & \swarrow & & & \swarrow & \downarrow & \swarrow & & \\ & & (a_1, c_1) & & & & & & (a_2, c_1) & & \end{array} $

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fong-spivak-seven-sketches_3 DIA_0498	330	—	—	—	
fong-spivak-seven-sketches_3 DIA_0499	331	—	—	—	
fong-spivak-seven-sketches_5 PIC_0001	5	—	—	—	
fong-spivak-seven-sketches_5 PIC_0002	5	—	—	—	
fong-spivak-seven-sketches_2 PIC_0003	29	—	—	—	$\emptyset$
fong-spivak-seven-sketches_1 PIC_0004	317	—	—	—	a) The map $[\text{id}_A + \iota_B, \iota_C] = [[\iota_A, \iota_B \circ \iota_{B+C}], \iota_C \circ \iota_{B+C}]: (A+B)+C \rightarrow A+(B+C)$ is an